

# $G(6, \mathbb{Z}_3\mathbb{C})$ Vacuum Structure and FFGFT

## Numerical Investigation of the Connection Questions

August 2026

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### Correction notes

**Corrections from the calculation review (1 Oct. 2026).** The following passages are corrected or annotated in the text; each carries a dated note with the previous value:

- Casimir equation in the section on Doc. 009: note that the QFT energy per area is  $-\pi^2 \hbar c / (720 a^3)$  without a factor  $4/3$ ; identifying it with  $C_2(SU(3)) = 4/3$  is an FFGFT interpretation [S].

## List of Symbols

### $G(6, \mathbb{Z}_3\mathbb{C})$ Symbols after Matzke (2026)

| Symbol                         | Meaning                                                                     | Remark                                                            |
|--------------------------------|-----------------------------------------------------------------------------|-------------------------------------------------------------------|
| $G(6, \mathbb{Z}_3\mathbb{C})$ | Clifford algebra, 6 dimensions, coefficient field $\{-1, 0, \pm 1, \pm i\}$ | $2 \equiv -1 \pmod{3}$                                            |
| $e_1, \dots, e_6$              | Basis vectors, $e_k^2 = +1$                                                 |                                                                   |
| $N_k^\pm$                      | Nilpotent vectors, $(N_k^\pm)^2 = 0$                                        | $N_k^\pm = e_a \pm i e_b$                                         |
| $P_{pk}$                       | Idempotents, $P_{pk}^2 = P_{pk}$                                            | $P_{pk} = (\mathbf{1} + i e_a e_b) / 2$                           |
| $V$                            | Vacuum operator, $V = N_1^+ N_2^+ N_3^+ N_1^- N_2^- N_3^-$                  | involution in $\mathbb{Z}_3\mathbb{C}$ : $V^2 \equiv -V \pmod{3}$ |
| $P_{Nk}$                       | Anti-particle idempotents, $P_{Nk}^2 = P_{Nk}$                              | $V = -64 \cdot P_{N1} P_{N2} P_{N3}$                              |
| $T_k$                          | Trine operator, $T_k^3 = \mathbf{1}$                                        | $T_k = \mathbf{1} + (\omega - 1) P_{pk}$                          |
| $\omega$                       | Primitive $\mathbb{Z}_3$ phase factor                                       | $e^{2\pi i / 3}$                                                  |
| $I, J, K$                      | Commutative quaternions (grade 4), $I^2 = J^2 = K^2 = +1$                   | encode 3 spatial dimensions                                       |
| $iE_7$                         | Complex volume form, $(iE_7)^2 = +1$                                        | universal phase operator                                          |

## FFGFT Symbols after Pascher (2023–2026)

| Symbol               | Meaning                                            | Value                                                    |
|----------------------|----------------------------------------------------|----------------------------------------------------------|
| $\xi$                | FFGFT geometry parameter                           | 4/30000                                                  |
| $T^4/\mathbb{Z}_3$   | 4D torus with $\mathbb{Z}_3$ orbifold projection   |                                                          |
| $\tau$               | Generator of the $\mathbb{Z}_3$ action on $T^4$    | unitary                                                  |
| $P_k$                | $\mathbb{Z}_3$ projectors on $L^2(T^4)$            | $\frac{1}{3} \sum_j \omega^{-jk} \tau^j$ , $k = 0, 1, 2$ |
| $\mathcal{H}_k$      | Eigensectors: $\mathcal{H}_k = P_k L^2(T^4)$       | three colour sectors                                     |
| $\hat{F}_{D_4}$      | $D_4$ sub-operator of the spectral operator        | $\xi = \lambda_{\min}(\hat{F}_{D_4})$                    |
| $C_2(R)$             | Quadratic Casimir operator (Lie algebra)           | $C_2(SU(3)_{\text{fund}}) = 4/3$                         |
| $N_{\text{Fourier}}$ | Fourier mode count of the $T^4$ torus              | $10^4 = 30000/ \mathbb{Z}_3 $                            |
| $N_c$                | Colour degrees of freedom, algebraically necessary | 3                                                        |
| $v$                  | Higgs vacuum expectation value                     | 246 GeV                                                  |

## FFGFT Document Numbering

This document references the FFGFT corpus (Johann Pascher, 2023–2026, ORCID 0009-0000-6518-4064):

- **Doc. XXX**: Document No. XXX of the FFGFT corpus. Full list: [github.com/jpascher/T0-Time-Mass-Duality](https://github.com/jpascher/T0-Time-Mass-Duality)
- **RXXX**: Correction/addendum No. XXX in the register Doc. 190.
- **[B]**: algebraically proved    **[K]**: derivable from  $\xi$     **[S]**: sketched/plausible

Documents relevant to this comparison: Doc. 009 (Casimir effect, factor 4/3), Doc. 049 (confinement), Doc. 160 ( $\alpha_s$ ), Doc. 314 ( $D_4$  lattice), Doc. 321 ( $SU(3)_c$  derivation), Doc. 322 (Hilbert space), Doc. 323 (Weinberg angle).

## Summary

This document compares the vacuum structure of  $G(6, \mathbb{Z}_3\mathbb{C})$  after Matzke [1] with FFGFT spectral theory and numerically investigates the open connection question: is  $\xi = 4/30000$  the smallest spectral value of the vacuum operator  $V$  in  $G(6)$ ?

Results of the numerical investigation (Python script 324\_G6\_FFGFT\_verify.py):

- The Trine theorem is verified:  $T_k^3 = \mathbf{1}$  to  $6.5 \times 10^{-16}$  **[B]**.
- Matzke's vacuum operator  $V = N_1^+ N_2^+ N_3^+ N_1^- N_2^- N_3^- = -64 \cdot P_{N_1} P_{N_2} P_{N_3}$  is an involution in  $\mathbb{Z}_3\mathbb{C}$ :  $V^2 = -64V$ , i.e.  $V^2 \equiv -V \pmod{3}$  **[B]**. The complementary Pp-product  $P_{P_1} P_{P_2} P_{P_3}$  is a rank-1 projector with eigenvalues  $\{0^{(7)}, 1^{(1)}\}$ , but lies on a different ray ( $|\langle \text{vac}_V | \text{vac}_{P_P} \rangle| = 0$ ).
- $\xi$  is *not* a direct spectral value of  $V$ . The connection is *structural*, not spectral.
- The trine product  $T_1 T_2 T_3$  is a global  $\mathbb{Z}_3$  operator:  $(T_1 T_2 T_3)^3 = \mathbf{1}$ , eigenvalues  $\{1^{(2)}, \omega^{(3)}, \omega^{2(3)}\}$  **[B]**.
- $\xi = C_2(SU(3)_{\text{fund}})/N_{\text{Fourier}} = (4/3)/10^4$  algebraically derived – R84 **[K]**.

### Status markers

**[K]** derivable from  $\xi$     **[B]** algebraically proved    **[S]** sketched/plausible

## .1 Starting Point: Two Frameworks, One Physics

### Matzke's $G(6, \mathbb{Z}_3\mathbb{C})$ Vacuum Structure

**[S]** Matzke constructs the physical vacuum state  $V$  as an algebraic object in  $G(6, \mathbb{Z}_3\mathbb{C})$ , a Clifford algebra over the coefficient field  $\mathbb{Z}_3\mathbb{C} = \{-1, 0, +1, \pm i\}$ . The central vacuum relation is

$$V^2 = -V \quad (\text{in } \mathbb{Z}_3\mathbb{C} \text{ arithmetic}), \quad (1)$$

where  $2 \equiv -1 \pmod{3}$ . Three nilpotent vector pairs  $N_k^\pm = e_a \pm i e_b$  with  $(N_k^\pm)^2 = 0$  decompose the  $\mathbb{C}^6$  spinor space into three eigensectors. The Trine theorem states: for every nilpotent  $N$  with  $N^2 = 0$ ,

$$T = \mathbf{1} + (\omega - 1)N \quad \Rightarrow \quad T^3 = \mathbf{1}. \quad (2)$$

### FFGFT Spectral Theory

**[K]** In FFGFT,  $\xi = 4/30000$  is the smallest eigenvalue of the  $D_4$  sub-operator  $\hat{F}_{D_4}$  (Doc. 322):

$$\xi = \lambda_{\min}(\hat{F}_{D_4}) = \frac{4}{30000}. \quad (3)$$

The  $\mathbb{Z}_3$  projection  $P_k = \frac{1}{3} \sum_{j=0}^2 \omega^{-jk} \tau^j$  on  $L^2(T^4)$  decomposes the Hilbert space into three eigensectors  $\mathcal{H}_0, \mathcal{H}_1, \mathcal{H}_2$ . Their bilinear transitions generate the  $\mathfrak{su}(3)$  algebra (Doc. 321, **[B]**).

## .2 Numerical Verification of the Trine Theorem

### Construction of the Vacuum Structure

We use the 8-dimensional spinor representation of  $G(6)$  over  $\mathbb{C}$ , given by  $\Gamma$  matrices as tensor products of Pauli matrices:

$$\begin{aligned} \Gamma_1 &= \sigma_x \otimes \mathbf{1} \otimes \mathbf{1}, & \Gamma_2 &= \sigma_y \otimes \mathbf{1} \otimes \mathbf{1}, & \Gamma_3 &= \sigma_z \otimes \sigma_x \otimes \mathbf{1}, \\ \Gamma_4 &= \sigma_z \otimes \sigma_y \otimes \mathbf{1}, & \Gamma_5 &= \sigma_z \otimes \sigma_z \otimes \sigma_x, & \Gamma_6 &= \sigma_z \otimes \sigma_z \otimes \sigma_y. \end{aligned} \quad (4)$$

The Clifford relation  $\{\Gamma_i, \Gamma_j\} = 2\delta_{ij}\mathbf{1}_8$  holds (numerically to  $< 10^{-14}$ ).

Nilpotents over three Witt pairs (0, 1), (2, 3), (4, 5):

$$N_k^+ = \Gamma_{2k} + i\Gamma_{2k+1}, \quad N_k^- = \Gamma_{2k} - i\Gamma_{2k+1}, \quad (N_k^\pm)^2 = 0 \quad \mathbf{[B]}. \quad (5)$$

Idempotents:

$$P_{Pk} = \frac{\mathbf{1} + i\Gamma_{2k}\Gamma_{2k+1}}{2}, \quad P_{Pk}^2 = P_{Pk} \quad (\text{numerically to } 0) \quad \mathbf{[B]}. \quad (6)$$

Anti-particle idempotents and vacuum operator:

$$P_{Nk} = \frac{\mathbf{1} - i\Gamma_{2k}\Gamma_{2k+1}}{2}, \quad P_{Nk}^2 = P_{Nk}, \quad P_{Pk}P_{Nk} = 0, \quad P_{Pk} + P_{Nk} = \mathbf{1} \quad \mathbf{[B]}. \quad (7)$$

$$V = N_1^+ N_2^+ N_3^+ N_1^- N_2^- N_3^- = -64 \cdot P_{N1} P_{N2} P_{N3}. \quad (8)$$

## Trine Operators and Their Spectrum

With  $\omega = e^{2\pi i/3}$ :

$$T_k = \mathbf{1} + (\omega - 1) P_{Pk}. \quad (9)$$

**[B]** Numerical result:

$$T_k^3 = \mathbf{1} \quad \text{to } 6.5 \times 10^{-16}. \quad (10)$$

The Trine theorem is exactly verified in the 8-dimensional spinor representation of  $G(6)$ .

The trine product:

$$(T_1 T_2 T_3)^3 = \mathbf{1} \quad \text{to } 1.9 \times 10^{-15}. \quad (11)$$

Eigenvalues of  $T_1 T_2 T_3$ :  $\{1^{(2)}, \omega^{(3)}, \omega^{2(3)}\}$  – a global  $\mathbb{Z}_3$  phase operator on the 8-dimensional spinor space.

## .3 Spectrum of the Vacuum Operator $V$

**[B]** Matzke's vacuum operator  $V = -64 \cdot P_{N1} P_{N2} P_{N3}$  has in the 8-dimensional spinor representation over  $\mathbb{C}$ :

$$\text{Tr}(V) = -64, \quad \text{rank}(V) = 1, \quad V^2 = -64 V. \quad (12)$$

In  $\mathbb{Z}_3\mathbb{C}$  arithmetic,  $-64 \equiv -1 \pmod{3}$ , hence:

$$V^2 \equiv -V \pmod{3} \quad (\text{involution — Matzke's relation confirmed [B]}). \quad (13)$$

The Pp-product  $P_{P1} P_{P2} P_{P3}$  is a rank-1 projector with  $\sigma = \{0^{(7)}, 1^{(1)}\}$ , but lies on a different ray:  $|\langle \text{vac}_V | \text{vac}_{Pp} \rangle| = 0$  **[B]**. The original analysis in Doc. 324 incorrectly identified the Pp-product with Matzke's  $V$ ; the correction is documented in `pruef_324_vakuum_involution.py`.

**[K]**  $\xi = 4/30000$  is *not* a spectral value of  $V$ . The connection is structural:

**Table 1:** Structural comparison  $G(6, \mathbb{Z}_3\mathbb{C})$  vs. FFGFT  $T^4/\mathbb{Z}_3$

| Quantity                | $G(6, \mathbb{Z}_3\mathbb{C})$                                    | FFGFT $T^4/\mathbb{Z}_3$                             |
|-------------------------|-------------------------------------------------------------------|------------------------------------------------------|
| $\mathbb{Z}_3$ symmetry | Witt decomposition of $\mathbb{C}^6$                              | Orbifold projection on $T^4$                         |
| Three sectors           | three nilpotent pairs $N_k^\pm$                                   | $\mathcal{H}_0, \mathcal{H}_1, \mathcal{H}_2$        |
| $N_c = 3$               | number of Witt pairs                                              | order of $\mathbb{Z}_3$                              |
| Confinement             | $T_R = 0$ (triality)                                              | $T_R = 0$ , algebraic [B]                            |
| Vacuum                  | involution $V = -64 \cdot P_{N1} P_{N2} P_{N3}$ , $V^2 \equiv -V$ | spectral value $\xi = \lambda_{\min}(\hat{F}_{D_4})$ |
| $SU(3)_c$               | bilinear transitions [S]                                          | proved [B] (Doc. 321)                                |
| Fermion masses          | nilpotents as directions                                          | eigenvalues $m_i = r_i \xi^{p_i} v$ [K]              |
| Numerator 4 in $\xi$    | 4 Witt pairs in $G(8)$                                            | 4 torus dimensions                                   |
| Denominator 30000       | $ \mathbb{Z}_3  \times 10^4$ (modes)                              | $ \mathbb{Z}_3  \times 10^4$ (modes)                 |

## .4 Agreements and Divergences

**[B]** The following structures are identical in both frameworks and numerically verified:

- Trine theorem:**  $T_k^3 = \mathbf{1}$  from nilpotents  $N_k^\pm$  – identical with the  $\mathbb{Z}_3$  phase structure of FFGFT projectors  $P_k = \frac{1}{3} \sum \omega^{-jk} \tau^j$ .
- Three sectors from one symmetry:**  $G(6)$  Witt decomposition  $\leftrightarrow$  FFGFT  $\mathbb{Z}_3$  eigenspace decomposition. Both yield exactly three sectors; their bilinear transitions generate  $\mathfrak{su}(3)$ .

3. **Confinement as  $T_R = 0$ :** Matzke's complement condition  $V(V + 1) = 0$  corresponds to the triality selection in Doc. 321.
4.  $N_c = 3$  **algebraically forced:** Neither  $G(6)$  nor FFGFT sets  $N_c$  as a free parameter.

**Table 2:** Open and closed bridges between  $G(6, \mathbb{Z}_3\mathbb{C})$  and FFGFT

| Point             | $G(6, \mathbb{Z}_3\mathbb{C})$ | FFGFT                             | Character     |
|-------------------|--------------------------------|-----------------------------------|---------------|
| Electroweak group | $SU(2)_L$ open [S]             | $SU(2)_L, U(1)_Y$ [B]             | G(6)-internal |
| Weinberg angle    | not computed                   | 0.2308 (−0.19 %) [K]              | G(6)-internal |
| Lepton masses     | not derived                    | $m_e, m_\mu, m_\tau < 1\%$ [K]    | G(6)-internal |
| Neutrino masses   | not derived                    | $m_{\nu_i}$ from fixed points [K] | G(6)-internal |
| $\xi$ connection  | Casimir quotient [K]           | $\xi = C_2/N_F$ [K]               | <b>closed</b> |

## .5 The Actual Connection Question

The question “is  $\xi$  the smallest spectral value of  $V$ ?” is fully answered by the numerical investigation and R84:

**[K]**  $\xi$  is not a spectral value of any  $G(6)$  operator – a closed negative finding, not an open bridge. The connection between  $G(6)$  and FFGFT lies on a different level:

$$\xi = \frac{C_2(SU(3)_{\text{fund}})}{N_{\text{Fourier}}} = \frac{4/3}{10^4} = \frac{4}{30000} \quad (14)$$

connects the *algebraic* structure of  $G(6)$  (Casimir invariant  $C_2 = 4/3$ , from  $N_c = 3$  [B]) with the *geometric* scale of FFGFT (torus mode count  $N_F = 10^4$ ). This is not a  $G(6)$ -internal statement, but the bridge formula between both frameworks – R84.

**Note on the electroweak group:**  $SU(2)_L \times U(1)_Y$  is algebraically derived in FFGFT (Doc. 321 [B]). Whether Matzke's  $G(6, \mathbb{Z}_3\mathbb{C})$  reproduces the same derivation internally is a  $G(6)$ -internal problem – not an open FFGFT bridge.

## .6 $\xi$ from the Casimir Operator of $SU(3)$

### The Quadratic Casimir Operator

**[B]** The quadratic Casimir operator of a Lie algebra  $\mathfrak{g}$  is the group invariant

$$C_2(R) = \sum_{a=1}^{\dim \mathfrak{g}} T_a^{(R)} T_a^{(R)}, \quad (15)$$

where  $T_a^{(R)}$  are the generators in representation  $R$ . For  $SU(N_c)$  in the *fundamental* representation:

$$C_2(SU(N_c)_{\text{fund}}) = \frac{N_c^2 - 1}{2N_c}. \quad (16)$$

With  $N_c = 3$  (algebraically necessary, Doc. 321, **[B]**):

$$C_2(SU(3)_{\text{fund}}) = \frac{9 - 1}{6} = \frac{8}{6} = \frac{4}{3}. \quad (17)$$

### $\xi$ as the Casimir Value per Fourier Mode

**[K]** The Fourier mode count of the  $T^4$  torus:

$$N_{\text{Fourier}} = \frac{N_{\text{modes}}}{|\mathbb{Z}_3|} = \frac{30000}{3} = 10^4. \quad (18)$$

The FFGFT geometry parameter follows:

$$\xi = \frac{C_2(SU(3)_{\text{fund}})}{N_{\text{Fourier}}} = \frac{4/3}{10^4} = \frac{4}{3 \times 10^4} = \frac{4}{30000} \quad (19)$$

**[B]** This formula is exact. Each quantity has a geometric or algebraic justification:

- $C_2(SU(3)_{\text{fund}}) = 4/3$ : algebraic from  $N_c = 3$ , which is algebraically necessary from the  $\mathbb{Z}_3$  structure (Doc. 321).
- $|\mathbb{Z}_3| = 3$ : order of the orbifold group (Doc. 321).
- $N_{\text{Fourier}} = 10^4$ : Fourier mode count of the  $T^4$  torus, from 4-dimensional topology and resolution scale.

### Connection to Doc. 009: Two Routes to the Factor 4/3

**[K]** Doc. 009 shows that the Casimir effect (vacuum energy between plates) delivers the factor 4/3 from QFT:

$$E_{\text{Casimir}} = -\frac{\pi^2 \hbar c}{720 a^3} \times \frac{4}{3}. \quad (20)$$

This is the same number as  $C_2(SU(3)_{\text{fund}}) = 4/3$  – not a coincidence. The physical Casimir effect (Doc. 009) is the field-theory manifestation of the algebraic Casimir value of the  $SU(3)$  algebra: in both cases 4/3 counts the degree-of-freedom weighting of an  $SU(3)$ -invariant system. *(Note of 1 Oct. 2026: The QFT Casimir energy per area between ideal plates is  $E/A = -\pi^2 \hbar c / (720 a^3)$  without a factor 4/3; the pressure is  $-\pi^2 \hbar c / (240 a^4)$  and the energy density  $-\pi^2 \hbar c / (720 a^4)$ , so the prefactor ratio pressure/energy density is 3, not 4/3 (check:  $|P| = 1.30 \text{ mPa}$  at  $a = 1 \mu\text{m}$ ). The factor 4/3 in the equation is appended; identifying it with  $C_2(SU(3)) = 4/3$  is an FFGFT interpretation **[S]**, not a QFT statement. Doc. 009 carries a corresponding note.)*

**Table 3:** Two routes to the factor 4/3 in  $\xi$

| Route                        | Source             | Factor                          |
|------------------------------|--------------------|---------------------------------|
| Casimir effect (physical)    | QFT, Doc. 009      | 4/3 from vacuum fluctuations    |
| Casimir operator (algebraic) | $SU(3)$ , Doc. 321 | $C_2 = (N_c^2 - 1)/(2N_c)$      |
| Result                       | both               | $4/3 \rightarrow \xi = 4/30000$ |

#### Casimir derivation of $\xi$ [K]

$\xi = C_2(SU(3)_{\text{fund}})/N_{\text{Fourier}}$  derives the numerator 4 in  $\xi = 4/30000$  algebraically from the quadratic Casimir operator.  $N_c = 3$  is algebraically necessary (Doc. 321 [B]); the factor 4/3 appears in both the algebraic Casimir and the physical Casimir effect (Doc. 009) – two independent routes to the same number.

## .7 Conclusion

### Overall status Doc. 324 – $G(6, \mathbb{Z}_3\mathbb{C}) \leftrightarrow \text{FFGFT}$

#### Verified [B]:

- Trine theorem:  $T_k^3 = \mathbf{1}$  in 8-dim spinor representation
- $V = -64 \cdot P_{N1}P_{N2}P_{N3}$  is an involution in  $\mathbb{Z}_3\mathbb{C}$ :  $V^2 \equiv -V \pmod{3}$ , rank = 1 [B]
- Trine product  $(T_1T_2T_3)^3 = \mathbf{1}$  with  $\mathbb{Z}_3$  spectrum
- Three sectors, bilinear transitions, confinement as  $T_R = 0$
- $C_2(SU(3)_{\text{fund}}) = 4/3$  algebraically from  $N_c = 3$

#### Closed [K]:

- $\xi$  is not a spectral value of  $V$  (numerically excluded) [K]
- $\xi = C_2(SU(3)_{\text{fund}})/N_{\text{Fourier}} = (4/3)/10^4 = 4/30000$  algebraically derived – R84 [K]
- The Laplace operator on the vacuum ray gives  $\Delta = 6 \cdot \mathbf{1}$  (trivial, since  $\Gamma_a^2 = \mathbf{1}$ ) – no connection to  $\xi$  as an eigenvalue; the question is closed as incorrectly posed [K]

#### Open [S]:

- $SU(2)_L \times U(1)_Y$  in  $G(6, \mathbb{Z}_3\mathbb{C})$  not yet derived – **G(6)-internal problem, not an FFGFT problem**. FFGFT has derived  $SU(2)_L$  and  $U(1)_Y$  from torus geometry (Doc. 321 [B]); the question is only whether Matzke's  $G(6, \mathbb{Z}_3\mathbb{C})$  reproduces the same derivation internally.

**Remark on the Laplace point:** The original question “is there a natural  $G(6)$  operator with  $\lambda_{\min} = \xi$ ?” is answered by R84:  $\xi$  does not arise from a spectrum within  $G(6)$ , but from the Casimir quotient  $C_2/N_{\text{Fourier}}$  – a bridge between algebraic structure ( $G(6)$ ) and geometric scale (FFGFT torus). These are two different levels; no  $G(6)$  operator alone can have  $\xi$  as an eigenvalue.

## References

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